GG-S-915B
21 September 1987
SUPERSEDING
GG-S-00915a(DSA-DM)
September 4, 1970
and GG-S-915
December 30, 1959

FEDERAL SPECIFICATION

SYRINGE, AIR, DENTAL and BULB, AIR SYRINGE, DENTAL, (Elastomer)

This Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a syringe and replacement bulb suitable for use with air in dental operative procedures.

1.2 Classification. This specification covers the following Types and Styles of items as specified (see 6.1):

Type I. Complete syringe.

Style 1. Valve in nozzle.

Style 2. Valve in bulb.

Type II. Replacement bulb.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawings. The following specifications, standards, and drawings of the issue in effect on date of invitation for bid or requests for proposal, form a part of this specification to the extent specified herein.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal Specifications

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiberboard,
PPP-B-676	Boxes, Setup.

Federal Standards

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-376	Preferred Metric Units for General Use by the Federal Government.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA).

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies).

Military Specifications

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-P-116	Methods of Preservation.

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Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

3. REQUIREMENTS

3.1 Material

3.1.1 Nozzle and Valves. The nozzle and valves shall be made of chrome plated brass or corrosion-resistant steel.

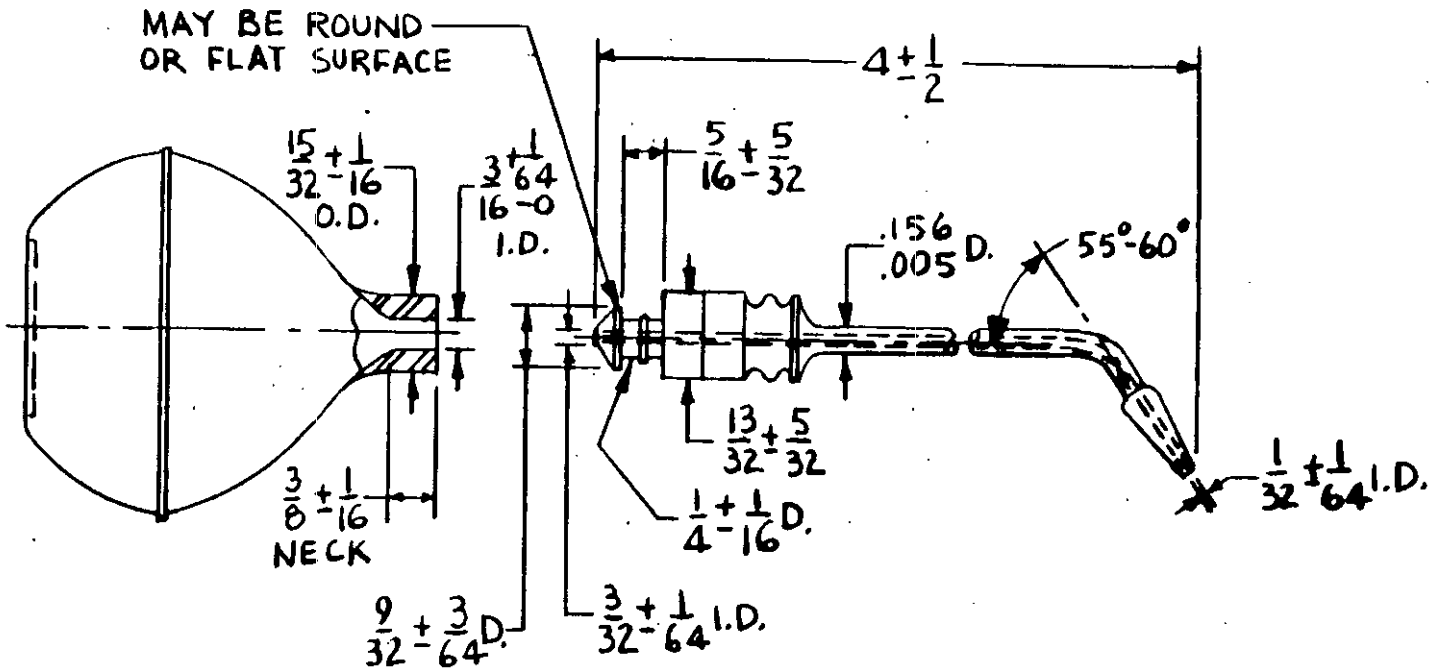
3.1.2 Bulb. Bulb shall be molded of a suitable elastomeric material (except for valve).

3.2 Design and construction.

3.2.1 Type I, Complete Syringe. The complete syringe shall be furnished as a style 1 or style 2. Type I, complete syringe shall be suitable for use and capable of directing a fine concentrated stream of air during dental operative procedures. Type I, Style 1 complete syringe shall conform in style, design, and dimensions to figure 1. Type I, Style 2 complete syringe shall conform in style, design, and dimensions to figure 2. The bulb and nozzle shall be easily connected and disconnected manually without the use of tools. The assembly, when joined, shall form an airtight seal.

3.2.1.1 Nozzle. The nozzle shall consist of a hollow tube stem, a bulbous base, and an optional overlapping ferrule on the proximal end. The conical tip of the nozzle shall be permanently attached. The nozzle of Type I, Style I, shall have a ball check air intake valve. The nozzle for Type I and Type II shall be designed to form an airtight union when fitted into the bulb (see 3.2.1.2).

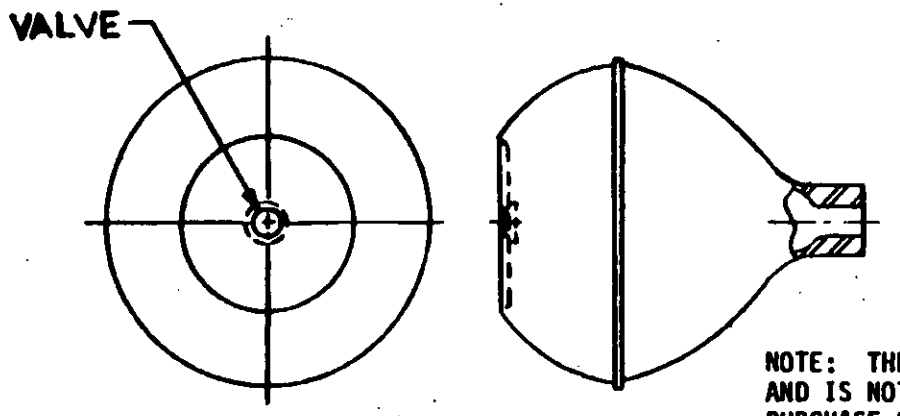
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TYPE I, STYLE I
COMPLETE SYRINGE

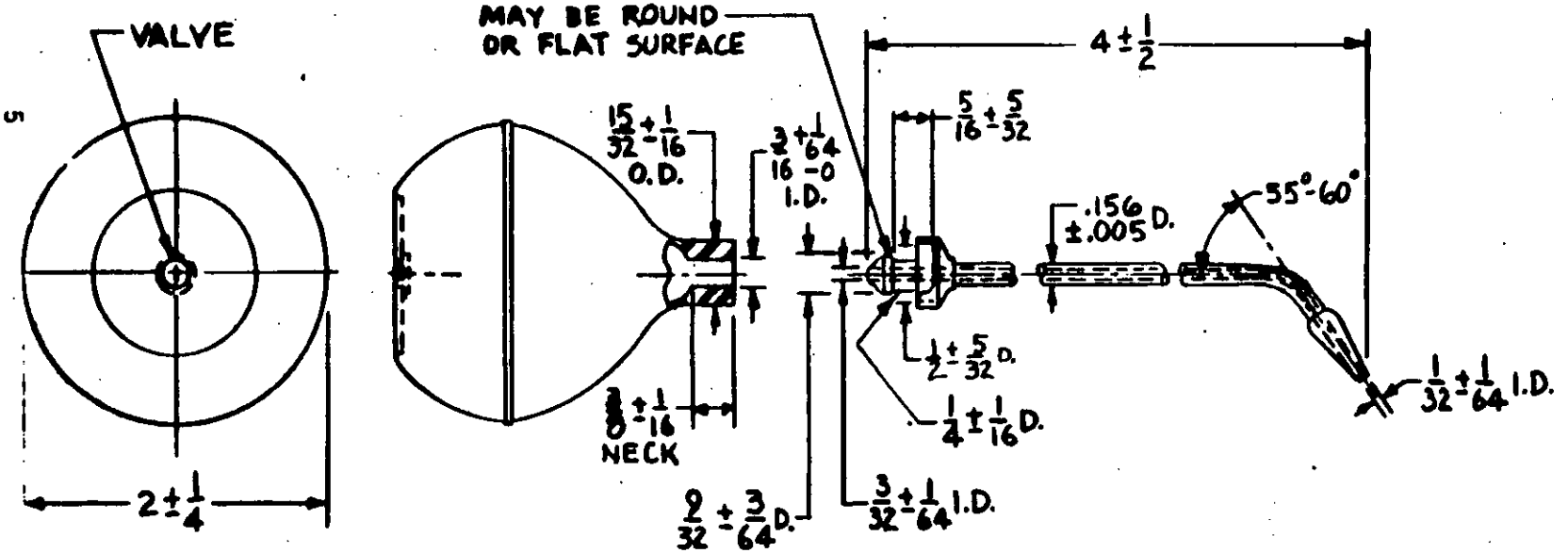
FIGURE 1. COMPLETE SYRINGE WITH VALVE IN NOZZLE

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TYPE II
REPLACEMENT BULB

NOTE: THE BULB SHOWN IS TYPICAL IN VALVE PLACEMENT AND IS NOT INTENDED TO RESTRICT OR PRECLUDE THE PURCHASE OF BULBS AND/OR SYRINGES WHICH ARE OTHERWISE IN ACCORDANCE WITH THE REQUIREMENTS OF THIS DOCUMENT.



TYPE I, STYLE II
COMPLETE SYRINGE

66-S-9158

FIGURE 2. COMPLETE SYRINGE WITH VALVE IN BULB AND REPLACEMENT BULB

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3.2.1.2 Bulb. The bulb shall have a nominal 2-ounce capacity. The bulb shall be round in shape with a flat base to facilitate an upright position. Bulb shall have a reinforced cylindrical neck with an opening bored through the center of the neck for reception of the nozzle. The bulb shall be easily compressed and shall be sufficiently resilient to serve its intended purpose. The bulb shall form an airtight union when fitted to the nozzle. The bulb of Type I, Style 2, complete syringe shall, in addition, have a ball check air intake valve.

3.2.2 Type II, replacement bulb. Type II replacement bulb shall conform to Type II of figure 2 for style, design, and dimensions, and shall be of the same style, design, and dimensions as the bulb section of Type I, Style 2 complete syringe.

3.3 Figures 1 and 2. Figures 1 and 2, which form a part of this specification, are specification control figures. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.4 Finish. Nozzles shall have a bright finish.

3.5 Corrosion resistance. Nozzles shall show no corrosion when subjected to the boil test as specified in 4.4.1. A slight corrosion in crevices shall not be cause for rejection.

3.6 Workmanship. Syringe and/or replacement bulbs shall show no pitting, crazing or checking and shall be free from defects which may detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Governments's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Inspection shall be conducted to determine compliance with specification requirements.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, elastomeric material, and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items to be offered for acceptance are classified as specified in Tables I, and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for inspection. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table I. Unit of product for examination purposes shall be one complete syringe for Type I and one replacement bulb for Type II.

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TABLE I. Inspection and Acceptable Quality Levels.

Type defect	Inspection Level	AQL (Percent Defective)
Major A	II	1.0
Major B	S-2	1.5
Major A	II	2.5
Major B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes.

TABLE II. Tests.

Characteristic	Requirement	Test procedure	Inspection level
Corrosion resistance	3.5	4.4.1	S-2

4.3 Examination. Complete syringes and replacement bulbs shall be examined for defects including, but not limited to, those listed in Table III.

TABLE III. Classification of Defects.

Categories	Defects
Major A:	
101	Type not as specified.
102	Style and design not as specified for applicable type.
103	Bulb not easily compressed.
104	Nozzle does not form airtight union with bulb.
105	Valve nonfunctional.
106	Conical tip not permanently attached.
107	Valve missing for applicable type and style.

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TABLE III. Classification of Defects (contuined).

Categories	Defects
Major B:	
151	Capacity of bulb not as specified.
152	Dimensions of nozzle affecting its fit into bulb not as specified.
153	Dimensions of hole bored through cylindrical neck of bulb not as specified.
154	Inside and outside diameters of nozzle not as specified.
155	Bulb base not flat.
156	Bulb and stem do not connect/disconnect easily without tools.
Minor A:	
201	Finish of nozzle not as specified.
202	Packaging and packing not as specified.
203	Marking not as specified.
Minor B:	
251	All other dimensions (other than those given in Major B defects) not as specified.

4.4 Tests.

4.4.1 Corrosion resistance. Syringe nozzle shall be subjected to the following boil test. Before testing, the nozzle shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the nozzle shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for an additional 24 hours.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. Packaging.

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Nozzles shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Nozzles shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative applications. Preservatives shall not be used.

5.1.1.4 Unit package.

5.1.1.4.1 Complete syringe Type I, Styles 1 and 2. Each complete syringe shall be packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification. As an alternate, complete syringe shall be packaged conforming to MIL-P-116, method IC-7.

5.1.1.4.2 Replacement bulb Type II. Each replacement bulb shall be packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-566, or PPP-B-676. Closure shall be as specified in the applicable box specification. As an alternate, replacement bulbs shall be packaged conforming to MIL-P-116, method IC-7.

5.1.2 Level C. The level C preservation for complete syringes and replacement bulbs shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Ninety-six unit packages containing complete syringes of one style only or replacement bulbs only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification.

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5.2.2 Level B. Ninety-six unit packages containing complete syringes of one style only or replacement bulbs shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for complete syringes or replacement bulbs shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 250 pounds or equal 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147.

Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 in height, 48 in width, and 40 inch length. Pallet shall conform to type IV or type V.

5.2.4.1 Levels A and B. Complete syringes or replacement bulbs packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Complete syringes or replacement bulbs packed as specified in 5.2.3, shall be unitized conforming to MIL-STD-794.

5.3 Marking.

5.3.1 Military activities.

5.3.1.1 Levels A, B and C. Each unit package, exterior container or unitized load shall be marked as specified in MIL-STD-129.

5.3.2 Civil agencies. Marking for shipment shall be as specified in FED-STD-123.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

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5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style desired (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Numbers	Item Identification
6520-00-501-8000	SYRINGE, AIR, DENTAL, BULB TYPE, 60ml
6520-00-501-9000	BULB, AIR SYRINGE, DENTAL, Elastomer

6.3 Subject term (key word) listing.

Air Syringe
Replacement Bulb

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

AGENT:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

47824

Project No. 6520-1837

Location: NCR Disk #10

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-S-915B	SYRINGE, AIR, DENTAL and BULB, AIR, SYRINGE, DENTAL (Elastomer)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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